

Clayton Glazier
03/23/2026
LIS 510
Mythbusting #4

"Two-Factor Authentication for Passwords is Overkill"

It is easy to understand why people believe that two-factor authentication is extra. Passwords have become a vital step in accessing digital profiles and data, such as ones laptop, bank account, gaming system, etc. Adding a second step to this process can feel unnecessary and annoying, as a strong password already feels like enough to stop others from accessing your stuff. The belief that our passwords are already strong enough is widespread, but it is a flawed assumption.

As Schneier explains in *Secrets and Lies*, passwords are fundamentally broken by design. They are supposed to be random enough to stop brute force attacks, while staying simple enough to remember. In practice, they are rarely both. People choose passwords from their government names or birthdays, or use simple variations across accounts. Even when systems store passwords, attackers can still access them using dictionary attacks that test millions of likely passwords very quickly. Imagine locking your front door but choosing a key shape that thousands of other people also use. It doesn't matter how good the lock is if the key isn't unique. A hacker can steal a database of passwords from a website and test those passwords across accounts very quickly.

Believing that passwords alone are sufficient leads individuals and organizations to rely on a single layer of defense. When that one single layer fails, access is given, and consequences can be severe. Bank account information, compromised emails, identity theft, and a danger for a future larger scale attack. Two factor authentication directly addresses this weakness. Instead of relying only on something you need to come up with and remember, it adds a second layer of defense, which is something you have. Even if an attacker cracks your password, they do not have your phone or laptop that is used to unlock the account in the second layer of defense.

Passwords are not enough on their own, no matter how complex they might seem. Schneier also talks about how attackers don't just break systems, they exploit predictable human behavior and computational power. The smart approach is using two-factor authentication to add another layer of defense. It is not overkill or unnecessary if it serves a real purpose in protection. It is the missing half of a security system that is sometimes not enough to stand on its own.